

Proof ledger

Repository source: proof/PROOF_LEDGER.md

Status labels:

- **Sourced:** stated and proved in an identified reference.
- **Hand-checked:** algebra reviewed in the project notes.
- **Exact-code checked:** a symbolic identity was verified over general symbolic parameters.
- **Numerically cross-checked:** deterministic quadrature/random tests agree.
- **Formal write-up drafted:** a line-by-line proof is in the repository, but has not received outside review.
- **Integrated manuscript drafted:** the claim has been merged into the LaTeX research note and compiled.
- **Hosted CI passed:** all configured GitHub Actions jobs completed successfully in a clean hosted environment.
- **Fresh-clone passed:** a newly cloned checkout ran all checks and builds and remained byte-clean.
- **Needs local rerun:** a matching check exists but should be rerun and preserved in the current repository state.
- **Needs external review:** no independent subject-matter expert has checked it.

ID	Claim	Current evidence	Status
N1	Support-function constraints and volume formula	Nishioka, Sections 2–4	Sourced
N2	Smooth approximation and volume continuity	Nishioka, Lemma 2; Martini–Montejano–Oliveros, Theorem 8.5.1; limiting argument in BRIDGE_LEMMAS.md	Sourced; formal write-up drafted
S1	Degree-3 coefficient is $1/22$; tail coefficient is at most $1/58$	Nishioka, Lemma 4 and Proposition 1; explicit derivation in BRIDGE_LEMMAS.md	Sourced; formal write-up drafted
S2	Tensor quadratic form is orthogonal across harmonic degrees	Polarized Bochner calculation in BRIDGE_LEMMAS.md, Lemmas 2.1–2.2	Hand-checked; formal write-up drafted; needs external review
T1	General cubic tensor is harmonic iff symmetric trace-free	DEGREE3_TENSOR_IDENTITY.md, Section 2; symbolic Laplacian check	Exact-code checked; formal write-up drafted; needs external review
T2	$ A_Y _2^2 = (176\pi/7)\tau$	Moment derivation in DEGREE3_TENSOR_IDENTITY.md, Section 5; general symbolic integration	Exact-code checked; formal write-up drafted; needs external review
T3	Exact formula for $ 4 \det A_Y _2^2$	Quartic contraction derivation in DEGREE3_TENSOR_IDENTITY.md, Sections 6–8; general symbolic integration	Exact-code checked by two formulations; formal write-up drafted; needs external review
T4	Compact pointwise determinant formula	DEGREE3_TENSOR_IDENTITY.md, Sections 3–4; matrix formula compared modulo $x^2+y^2+z^2-1$	Exact-code checked; hand-checked; needs external review

ID	Claim	Current evidence	Status
T5	Rotational invariance and random convention checks	Deterministic sphere quadrature and random rotations	Numerically cross-checked independently in Python and R; both earlier runs pass
T6	Pairing counts for the six quartic moments	Hand count in <code>DEGREE3_TENSOR_IDENTITY.md</code> , Section 7; exhaustive exact pairing enumeration	Exact-count checks passed independently in Python and R
F1	Nuclear-norm identity and Cauchy–Schwarz step	<code>BRIDGE_LEMMAS.md</code> , Lemmas 4.1–4.2	Hand-checked; formal write-up drafted; needs external review
F2	Projection/duality step $\ C\ _2^2 = \langle A_h, C \rangle \leq (d/2) \int \ C\ _*$	<code>BRIDGE_LEMMAS.md</code> , Corollary 2.3 and Lemmas 5.1–5.2	Hand-checked; formal write-up drafted; needs external review
A1	Final exact coefficient, decimal, and strict improvement	SymPy exact arithmetic; <code>BRIDGE_LEMMAS.md</code> , Theorem 6.2 and Corollary 6.3	Exact-code checked; hand-checked; integrated manuscript drafted
R1	Passage to nonsmooth bodies	Nishioka’s approximation lemma; <code>BRIDGE_LEMMAS.md</code> , Theorem 7.1	Sourced; formal write-up drafted; needs external review
P1	Complete paper source compiles without unresolved references	<code>paper/degree3_spectral_refinement.tex</code> ; <code>build_paper.sh</code>	Integrated manuscript drafted; runtime-tested in project environment
P2	PDF has no visible clipping, overlap, or broken glyphs	Eight rendered pages inspected at 180 dpi	Render-checked
C1	Hosted clean-environment checks for Python, R, the paper, metadata, and Markdown PDFs	<code>.github/workflows/verification.yml</code> ; three green jobs confirmed for the tested commit on September 4, 2026	Hosted CI passed
C2	Fresh-clone reproduction of all checks and tracked builds	<code>CLEAN_CLONE_CHECK.md</code> ; commit <code>96cea5727507</code>	Fresh-clone passed; rebuilt tracked tree byte-clean
D1	Public-facing verification and reproduction instructions	<code>VERIFICATION_STATUS.md</code> ; <code>REPRODUCIBILITY.md</code>	Drafted and integrated
M1	Authorship, AI provenance, citation, and licensing metadata	<code>AI_ASSISTANCE.md</code> ; <code>CITATION.cff</code> ; <code>LICENSE</code> ; manuscript declaration; metadata preflight	Drafted and integrated; public release URL/version pending
L1	Exact constant/decimal not found in a targeted web search	Search terms and limitations recorded in <code>PROOF_AUDIT.md</code>	Preliminary only; not a systematic novelty review
E1	Full theorem has been independently reviewed	None yet	Needs external review

Current bottleneck

The internal proof chain, integrated manuscript, hosted verification workflow, authorship/provenance metadata, and an initial clean-clone reproduction are complete. The remaining internal gates are:

1. have the named author finish reading and approve the manuscript;
2. complete the release-candidate audit and rerun the clean-clone check after release-specific metadata is fixed; and

3. add the public repository URL, release version, and release date to `CITATION.cff` when the first public release is prepared.

Release rule

Do not describe the candidate bound as peer reviewed, certified, or independently verified while **E1** remains open. The first public GitHub release may be labeled **public research draft seeking verification** after all internal gates above pass.